Companion Results Appendix: GGUF Throughput Prediction

Protocol-complete two-host evidence for the ICASSP manuscript

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1 Scope and cohort

The source manifest contains 23 GGUF files. Across two hosts, the scored data contain 21 unique files and 33 host–file configurations (27 training and 6 held out). Each configuration has one decode and one prefill observation at each of the 3 measured context depths, for 99 decode and 99 prefill rows. The final protocol selector retains 216 observations from 22 unique files in total: 198 scored rows and 18 reference-probe rows excluded from prediction fits and scores. Probe exclusion is host-specific, so a file used as a probe on one host can be scored on the other. All exported rows carry the declared protocol fields. The retry decision, however, uses the worst within-run *decode* CV in an invocation. Prefill rows inherit that invocation's protocol metadata but are not gated on their own CV. This distinction matters on RTX 5080, where the noisiest prefill row has 39.19% CV even though the associated decode-gate CV is 1.35%.

Host	Backend	Train cfg.	Test cfg.	Scored decode	Scored prefill
Apple M4 Max	BLAS,MTL	15	4	57	57
RTX 5080	CUDA	12	2	42	42

Host	Retried cells	Gate CV med.	Gate CV max	PP CV med.	PP CV max	PP > 3%
Apple M4 Max	3	1.43	3.04	1.41	4.84	7/57
RTX 5080	3	0.88	1.65	3.56	39.19	29/42

Attempted configurations with no successful row are Apple M4 Max: `gpt-oss-120b-MXFP4.gguf`. On Apple M4 Max, two raw attempts of the one gpt-oss-120B configuration report `failed to decode prompt batch, res=-3`; these are prompt-batch failures, not demonstrated model-load or out-of-memory failures. Host-specific calibration probes are Apple M4 Max: `Qwen3.8-27B-Q8_0.gguf`, `Qwen3.5-9B-Q8_0.gguf`, `Qwen3.5-4B-Q8_0.gguf`; RTX 5080: none.

1.1 Cohort accounting

Raw records, superseded legacy rows, repeated protocol attempts, selected observations, and scoring exclusions are separated below. A protocol duplicate is a successful protocol row removed by the atomic host–model–phase–depth selector; it is not an additional scored observation.

Host	Raw	Failed	Legacy ok	Protocol ok	Dup. removed	Selected	Probes	Scored
Apple M4 Max	278	2	132	144	12	132	18	114
RTX 5080	84	0	0	84	0	84	0	84

1.2 Excluded reference-probe observations

These rows complete the 216-observation selected cohort and make the descriptive matched-host and quantization comparisons reproducible. They are reported only as measurements, never as prediction errors.

Table 1: Selected protocol-complete reference-probe observations excluded from prediction fitting and scoring. Measured throughput and SD are in tokens/s; IDs resolve through Table 2.

Host	ID	Phase	Depth	Measured	SD	CV %	Attempts
Apple M4 Max	M09	decode	0	77.275	0.201	0.260	1
Apple M4 Max	M09	decode	4,096	74.687	0.153	0.204	1

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Table 1: Selected protocol-complete reference-probe observations excluded from prediction fitting and scoring. Measured throughput and SD are in tokens/s; IDs resolve through Table 2. (continued)

Host	ID	Phase	Depth	Measured	SD	CV %	Attempts
Apple M4 Max	M09	decode	16,384	67.270	0.341	0.507	1
Apple M4 Max	M09	prefill	0	1553.086	1.682	0.108	1
Apple M4 Max	M09	prefill	4,096	1416.098	1.799	0.127	1
Apple M4 Max	M09	prefill	16,384	992.403	10.398	1.048	1
Apple M4 Max	M11	decode	0	46.499	0.380	0.818	2
Apple M4 Max	M11	decode	4,096	45.340	0.169	0.372	2
Apple M4 Max	M11	decode	16,384	42.119	0.938	2.226	2
Apple M4 Max	M11	prefill	0	865.397	0.661	0.076	2
Apple M4 Max	M11	prefill	4,096	784.470	4.299	0.548	2
Apple M4 Max	M11	prefill	16,384	560.769	14.295	2.549	2
Apple M4 Max	M13	decode	0	14.902	0.054	0.365	1
Apple M4 Max	M13	decode	4,096	14.433	0.017	0.118	1
Apple M4 Max	M13	decode	16,384	13.575	0.092	0.680	1
Apple M4 Max	M13	prefill	0	238.511	7.257	3.043	1
Apple M4 Max	M13	prefill	4,096	201.115	2.027	1.008	1
Apple M4 Max	M13	prefill	16,384	161.599	1.227	0.759	1

Probe-exclusion sensitivity. The three Mac Q8 files defined the historical LLM-reference diagnostic. Their exclusion was fixed before the final device-copy analysis and is retained to avoid a post-hoc cohort change. If they are included and B2 is refit, MAPE is 10.46% over 54 training rows and 13.11% over 12 held-out rows; the latter is unchanged because all probes are Q8 training files while the test files are Q4. On the original 45 non-probe training rows under this refit, MAPE is 11.06%.

2 Aggregate prediction results

Results are separated by host because fitted efficiency factors are host-specific; a pooled row would weight the unequal host cohorts and is not a cross-system score.

Host	Predictor	Split	n	MAPE (%)	Median (%)	P90 / max (%)
Apple M4 Max	B0 unfitted (eta=1, total)	train	45	33.57	19.37	76.47 / 123.27
Apple M4 Max	B0 unfitted (eta=1, total)	test	12	46.89	49.97	83.45 / 84.22
Apple M4 Max	B1 fitted, total params	train	45	15.80	3.20	70.29 / 76.77
Apple M4 Max	B1 fitted, total params	test	12	49.36	54.66	81.90 / 82.74
Apple M4 Max	B2 fitted, active params (ours)	train	45	10.39	2.12	26.32 / 78.49
Apple M4 Max	B2 fitted, active params (ours)	test	12	13.11	8.98	36.65 / 49.92
RTX 5080	B0 unfitted (eta=1, total)	train	36	20.27	14.47	44.15 / 71.19
RTX 5080	B0 unfitted (eta=1, total)	test	6	41.65	40.39	81.22 / 81.95
RTX 5080	B1 fitted, total params	train	36	9.85	8.88	20.95 / 31.73
RTX 5080	B1 fitted, total params	test	6	51.85	50.81	84.50 / 85.11
RTX 5080	B2 fitted, active params (ours)	train	36	10.12	9.34	20.95 / 31.73
RTX 5080	B2 fitted, active params (ours)	test	6	36.15	25.79	62.41 / 68.96
Apple M4 Max	P1 eta_p per host	train	15	6.81	5.36	16.58 / 33.40
Apple M4 Max	P1 eta_p per host	test	4	22.68	19.16	44.95 / 51.25
Apple M4 Max	P2 eta_p per host x quant	train	15	4.13	0.00	15.06 / 25.99
Apple M4 Max	P2 eta_p per host x quant	test	4	18.68	14.90	36.90 / 42.85
RTX 5080	P1 eta_p per host	train	12	14.46	9.15	31.38 / 52.16
RTX 5080	P1 eta_p per host	test	2	124.51	124.51	207.24 / 227.93
RTX 5080	P2 eta_p per host x quant	train	12	5.86	0.00	21.31 / 25.36
RTX 5080	P2 eta_p per host x quant	test	2	108.18	108.18	184.89 / 204.07

Decode summaries use all three context depths. Prefill headline rows use only depth 0, because the stated prefill model has no existing-prefix term; deeper rows are scope diagnostics below.

2.1 Leave-one-host-out transfer

B2 transfers the source host's median per-quantization coefficients and substitutes the untouched target host's bandwidth. The target set contains all eligible rows on that host; because most files also occur on the source host, this isolates host/runtime transfer rather than a simultaneous model-and-hardware holdout. Q4_K_M is shared; the Mac Q4_K test file uses the source-wide median fallback. The second variant is the rejected absolute-time output-projection extension.

Variant	Target host	Source n	Target n	MAPE	Median	Max
B2: all target rows	Apple M4 Max	36	57	20.82	15.41	75.81
B2: all target rows	RTX 5080	45	42	21.69	18.75	71.53
B2: target test rows	Apple M4 Max	36	12	13.94	8.63	47.67
B2: target test rows	RTX 5080	45	6	36.70	26.18	71.53
rejected two-term	Apple M4 Max	36	57	57.06	54.22	78.57
rejected two-term	RTX 5080	45	42	116.85	98.66	349.25

For context, target-fitted B2 gives all-row MAPE of 10.96% on Mac and 13.84% on RTX. Unlike the untouched-target transfer rows, these comparators mix training rows used to fit the target coefficients with held-out rows.

Host	Split	Architecture	n	B0	B1	B2
Apple M4 Max	train	dense	36	23.21	7.14	5.66
Apple M4 Max	train	MoE	9	74.98	50.45	29.29
Apple M4 Max	test	dense	6	11.18	17.75	3.28
Apple M4 Max	test	MoE	6	82.61	80.97	22.95
RTX 5080	train	dense	33	15.73	10.57	10.57
RTX 5080	train	MoE	3	70.28	1.95	5.18
RTX 5080	test	dense	3	3.54	20.40	20.40
RTX 5080	test	MoE	3	79.77	83.31	51.90

2.2 Per-layer KV correction

With each variant refit on training rows, replacing a uniform-global KV denominator by the frozen per-layer metadata produces:

Host	Split	n	Uniform KV MAPE (%)	Per-layer KV MAPE (%)
Apple M4 Max	train	45	12.68	10.39
Apple M4 Max	test	12	21.23	13.11
RTX 5080	train	36	12.37	10.12
RTX 5080	test	6	37.87	36.15

2.3 Decode context behavior

All 33 scored host–file configurations slow monotonically from 0 through 16,384 tokens. The normalized rate is each model's throughput divided by its own depth-0 throughput.

Host	Depth	Train n	Test n	Train B2 MAPE	Test B2 MAPE	Median normalized rate
Apple M4 Max	0	15	4	11.30	17.86	1.000
Apple M4 Max	4,096	15	4	7.59	12.83	0.945
Apple M4 Max	16,384	15	4	12.27	8.63	0.837
RTX 5080	0	12	2	10.19	44.52	1.000
RTX 5080	4,096	12	2	4.49	38.28	0.985
RTX 5080	16,384	12	2	15.67	25.64	0.922

2.4 Prefill scope diagnostic

Depth 0 is the primary prefill fit and score. The same fitted coefficients are applied unchanged at larger existing-prefix depths. These errors are prediction errors, distinct from within-run prefill variability; the RTX held-out result is poor even before that distinction is considered.

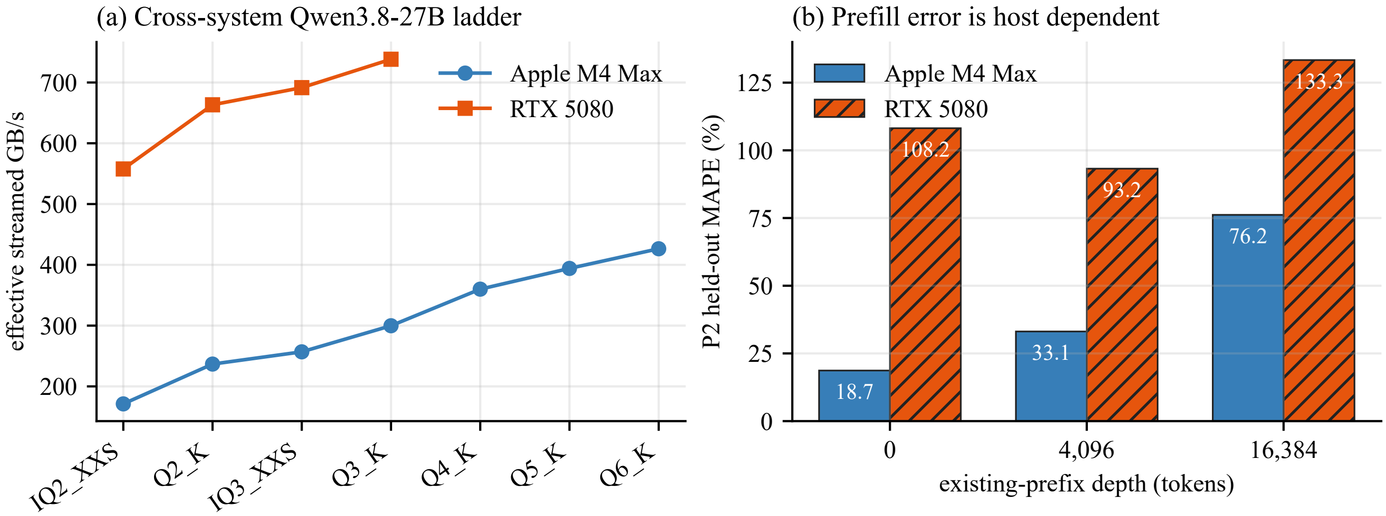

Figure 1: Host-separated diagnostics. Left: effective streamed bandwidth for the scored Qwen3.8-27B quantizations at zero prefix; host coverage differs, and IQ2 has 64 layers/26.90B parameters versus 65 layers/27.32B for the other shown files, so the set is a family comparison rather than a strictly identical-network quantization ladder. Right: P2 held-out error when depth-0 coefficients are applied at each existing-prefix depth. The RTX result is not pooled with the Apple result.

Host	Split	Depth	*n*	P1 MAPE	P2 MAPE	P2 median	P2 max
Apple M4 Max	train	0	15	6.81	4.13	0.00	25.99
Apple M4 Max	train	4,096	15	22.35	20.83	17.58	49.82
Apple M4 Max	train	16,384	15	66.94	64.13	51.02	150.97
Apple M4 Max	test	0	4	22.68	18.68	14.90	42.85
Apple M4 Max	test	4,096	4	40.98	33.11	29.60	55.36
Apple M4 Max	test	16,384	4	86.59	76.18	73.78	132.42
RTX 5080	train	0	12	14.46	5.86	0.00	25.36
RTX 5080	train	4,096	12	19.26	12.35	8.23	32.74
RTX 5080	train	16,384	12	36.13	28.56	18.19	72.26
RTX 5080	test	0	2	124.51	108.18	108.18	204.07
RTX 5080	test	4,096	2	108.39	93.23	93.23	167.57
RTX 5080	test	16,384	2	151.64	133.34	133.34	227.36

3 Selected model metadata

Tables 2 and 3 jointly reproduce every frozen metadata field used for the 22 unique selected files (21 scored and 1 probe-only). Scored host coverage is explicit because the two measurement cohorts are not identical.

Table 2: Selected model identity, source, and file size. Sizes are exact bytes; host codes are M4 (Apple M4 Max) and RTX (RTX 5080), and "probe only" denotes a file excluded from every prediction fit and score.

ID	Split	Scored host(s)	GGUF filename	Repository	Architecture	Quant.	Bytes
M01	train	M4, RTX	gemma-4-12b-it-qat-q4_0.gguf	google/gemma-4-12B-it-qat-q4_0-gguf	gemma4	Q4_0	6,975,879,296
M02	train	M4	gemma-4-26B-A4B-it-UD-Q4_K_M.gguf	unsloth/gemma-4-26B-A4B-it-GGUF	gemma4	Q4_K_M	16,947,541,728
M03	train	M4, RTX	gpt-oss-20b-MXFP4.gguf	ggml-org/gpt-oss-20b-GGUF	gpt-oss	MXFP4	12,109,566,624
M04	test	M4, RTX	inclusionAI_Ling-mini-2.0-Q4_K_M.gguf	bartowski/inclusionAI_Ling-mini-2.0-GGUF	bailingmoe2	Q4_K_M	9,941,894,336
M05	test	M4	Muse-Glimmer-30B-UD-Q4_K_XL.gguf	unsloth/Muse-Glimmer-30B-GGUF	muse-glimmer	Q4_K	15,878,222,368
M06	test	M4	Nemotron-3-Nano-30B-A3B-Q4_K_M.gguf	unsloth/Nemotron-3-Nano-30B-A3B-GGUF	nemotron_h_moe	Q4_K_M	24,574,373,664
M07	test	M4, RTX	NVIDIA-Nemotron-3-Nano-4B-Q4_K_M.gguf	lmstudio-community/NVIDIA-Nemotron-3-Nano-4B-GGUF	nemotron_h	Q4_K_M	2,837,072,896
M08	train	M4, RTX	Qwen3.5-4B-Q4_K_M.gguf	unsloth/Qwen3.5-4B-GGUF	qwen35	Q4_K_M	2,740,937,888
M09	train	RTX	Qwen3.5-4B-Q8_0.gguf	unsloth/Qwen3.5-4B-GGUF	qwen35	Q8_0	4,482,403,488
M10	train	M4, RTX	Qwen3.5-9B-Q4_K_M.gguf	unsloth/Qwen3.5-9B-GGUF	qwen35	Q4_K_M	5,680,522,464
M11	train	RTX	Qwen3.5-9B-Q8_0.gguf	unsloth/Qwen3.5-9B-GGUF	qwen35	Q8_0	9,527,502,048
M12	train	M4	Qwen3.6-35B-A3B-UD-Q4_K_M.gguf	unsloth/Qwen3.6-35B-A3B-GGUF	qwen35moe	Q4_K_M	22,134,528,992
M13	probe only	–	Qwen3.8-27B-Q8_0.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q8_0	29,047,086,048
M14	train	M4, RTX	Qwen3.8-27B-UD-IQ2_XXS.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	IQ2_XXS	7,266,070,528
M15	train	M4, RTX	Qwen3.8-27B-UD-IQ3_XXS.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	IQ3_XXS	10,934,860,704
M16	train	M4, RTX	Qwen3.8-27B-UD-Q2_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q2_K	9,828,981,664
M17	train	M4, RTX	Qwen3.8-27B-UD-Q3_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q3_K	13,146,393,504
M18	train	M4	Qwen3.8-27B-UD-Q4_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q4_K	17,559,178,144
M19	train	M4	Qwen3.8-27B-UD-Q5_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q5_K	20,876,938,144
M20	train	M4	Qwen3.8-27B-UD-Q6_K_XL.gguf	unsloth/Qwen3.8-27B-GGUF	qwen35	Q6_K	25,299,061,664
M21	train	M4, RTX	SmolLM3-Q4_K_M.gguf	ggml-org/SmolLM3-3B-GGUF	smollm3	Q4_K_M	1,915,305,312
M22	train	M4, RTX	SmolLM3-Q8_0.gguf	ggml-org/SmolLM3-3B-GGUF	smollm3	Q8_0	3,275,574,624

Table 3: Selected model geometry. KV columns are exact bytes at the measured depths; k/E is routed/total experts.

ID	Total P	Active P	Active %	Layers	Width	Vocab.	KV heads	Head dim.	Used/all	KV@0	KV@4096	KV@16384
M01	11,907,350,576	11,907,350,576	100.0	48	3,840	262,144	8	512	–	0	738,197,504	939,524,096
M02	25,233,142,046	3,822,527,246	15.1	30	2,816	262,144	8	512	8/128	0	503,316,480	754,974,720
M03	20,914,757,184	4,187,440,704	20.0	24	2,880	201,088	8	64	4/32	0	201,326,592	805,306,368
M04	16,255,643,392	1,432,973,056	8.8	20	2,048	157,184	4	128	8/256	0	167,772,160	671,088,640
M05	27,854,794,240	27,854,794,240	100.0	52	6,656	202,048	2	128	–	0	218,103,808	872,415,232
M06	31,577,940,288	3,580,076,352	11.3	52	2,688	131,072	2	128	6/128	397,934,592	423,100,416	498,597,888
M07	3,973,556,832	3,973,556,832	100.0	42	3,136	131,072	8	128	–	616,366,080	683,474,944	884,801,536
M08	4,205,751,296	4,205,751,296	100.0	32	2,560	248,320	4	256	–	0	536,870,912	2,147,483,648
M09	4,205,751,296	4,205,751,296	100.0	32	2,560	248,320	4	256	–	0	536,870,912	2,147,483,648
M10	8,953,803,264	8,953,803,264	100.0	32	4,096	248,320	4	256	–	0	536,870,912	2,147,483,648
M11	8,953,803,264	8,953,803,264	100.0	32	4,096	248,320	4	256	–	0	536,870,912	2,147,483,648
M12	34,660,610,688	3,454,988,928	10.0	40	2,048	248,320	2	256	8/256	0	335,544,320	1,342,177,280
M13	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M14	26,895,998,464	26,895,998,464	100.0	64	5,120	248,320	4	256	–	0	1,073,741,824	4,294,967,296
M15	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M16	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M17	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M18	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M19	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M20	27,320,697,856	27,320,697,856	100.0	65	5,120	248,320	4	256	–	0	1,090,519,040	4,362,076,160
M21	3,075,098,624	3,075,098,624	100.0	36	2,048	128,256	4	128	–	0	301,989,888	1,207,959,552
M22	3,075,098,624	3,075,098,624	100.0	36	2,048	128,256	4	128	–	0	301,989,888	1,207,959,552

4 All scored decode observations

These are the 99 rows exported by `results/predictions.csv`. Throughput is tokens/s; APE is absolute percentage error. Rows are ordered by host, split, model ID, and depth.

Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts.

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
Apple M4 Max	M01	train	0	54.28	54.28	0.00	0.42	15.79	1
Apple M4 Max	M01	train	4,096	49.91	49.08	1.65	0.55	15.79	1
Apple M4 Max	M01	train	16,384	45.34	47.84	5.50	1.54	15.79	1
Apple M4 Max	M02	train	0	89.15	124.01	39.11	0.15	16.58	1
Apple M4 Max	M02	train	4,096	82.91	103.69	25.07	0.42	16.58	1
Apple M4 Max	M02	train	16,384	75.37	95.83	27.15	0.44	16.58	1
Apple M4 Max	M03	train	0	124.28	128.17	3.13	0.29	11.83	1
Apple M4 Max	M03	train	4,096	118.34	118.34	0.00	0.19	11.83	1
Apple M4 Max	M03	train	16,384	102.83	96.21	6.43	0.67	11.83	1
Apple M4 Max	M08	train	0	101.64	116.16	14.29	0.16	8.19	1
Apple M4 Max	M08	train	4,096	97.13	97.13	0.00	0.28	8.19	1
Apple M4 Max	M08	train	16,384	85.05	65.13	23.42	1.25	8.19	1
Apple M4 Max	M10	train	0	66.64	56.05	15.90	1.00	8.30	1
Apple M4 Max	M10	train	4,096	62.95	51.21	18.65	1.81	8.30	1
Apple M4 Max	M10	train	16,384	56.62	40.67	28.17	2.81	8.30	1
Apple M4 Max	M12	train	0	80.85	144.30	78.49	0.29	15.62	1
Apple M4 Max	M12	train	4,096	78.38	125.26	59.81	0.24	15.62	1
Apple M4 Max	M12	train	16,384	72.10	89.72	24.44	1.12	15.62	1
Apple M4 Max	M14	train	0	23.56	23.56	0.00	1.20	16.52	1
Apple M4 Max	M14	train	4,096	20.41	20.53	0.59	2.70	16.52	1
Apple M4 Max	M14	train	16,384	19.71	14.81	24.88	1.72	16.52	1
Apple M4 Max	M15	train	0	23.49	24.42	3.94	1.09	14.84	1
Apple M4 Max	M15	train	4,096	22.20	22.20	0.00	1.59	14.84	1
Apple M4 Max	M15	train	16,384	19.63	17.45	11.07	1.65	14.84	1
Apple M4 Max	M16	train	0	24.08	25.71	6.75	1.90	11.04	1
Apple M4 Max	M16	train	4,096	23.14	23.14	0.00	2.39	11.04	1
Apple M4 Max	M16	train	16,384	21.30	17.81	16.41	1.65	11.04	1
Apple M4 Max	M17	train	0	22.80	23.29	2.12	2.60	11.42	3
Apple M4 Max	M17	train	4,096	21.89	21.50	1.75	0.91	11.42	3
Apple M4 Max	M17	train	16,384	17.49	17.49	0.00	2.94	11.42	3
Apple M4 Max	M18	train	0	20.50	20.92	2.05	2.11	13.06	3

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Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts. (continued)

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
Apple M4 Max	M18	train	4,096	19.70	19.70	0.00	1.97	13.06	3
Apple M4 Max	M18	train	16,384	16.77	16.76	0.10	1.47	13.06	3
Apple M4 Max	M19	train	0	18.88	19.13	1.31	1.57	17.66	1
Apple M4 Max	M19	train	4,096	18.33	18.18	0.83	2.21	17.66	1
Apple M4 Max	M19	train	16,384	15.82	15.82	0.00	2.67	17.66	1
Apple M4 Max	M20	train	0	16.86	16.86	0.00	0.37	17.18	1
Apple M4 Max	M20	train	4,096	15.92	16.16	1.50	1.42	17.18	1
Apple M4 Max	M20	train	16,384	14.97	14.38	3.96	0.92	17.18	1
Apple M4 Max	M21	train	0	169.50	166.23	1.93	0.46	13.91	1
Apple M4 Max	M21	train	4,096	149.47	143.59	3.93	0.31	13.91	1
Apple M4 Max	M21	train	16,384	115.22	101.94	11.53	0.40	13.91	1
Apple M4 Max	M22	train	0	118.87	119.53	0.55	0.40	13.92	1
Apple M4 Max	M22	train	4,096	109.44	109.44	0.00	0.41	13.92	1
Apple M4 Max	M22	train	16,384	88.22	87.32	1.01	1.00	13.92	1
Apple M4 Max	M04	test	0	242.33	363.29	49.92	1.24	11.64	1
Apple M4 Max	M04	test	4,096	219.67	304.92	38.81	0.84	11.64	1
Apple M4 Max	M04	test	16,384	175.60	205.75	17.17	0.37	11.64	1
Apple M4 Max	M05	test	0	26.23	23.14	11.81	1.92	8.93	3
Apple M4 Max	M05	test	4,096	23.30	22.82	2.05	2.13	8.93	3
Apple M4 Max	M05	test	16,384	22.55	21.93	2.74	3.03	8.93	3
Apple M4 Max	M06	test	0	92.12	100.00	8.55	0.30	10.01	1
Apple M4 Max	M06	test	4,096	90.68	99.21	9.41	0.49	10.01	1
Apple M4 Max	M06	test	16,384	85.16	96.93	13.83	0.68	10.01	1
Apple M4 Max	M07	test	0	93.29	92.19	1.18	0.30	8.78	1
Apple M4 Max	M07	test	4,096	91.41	90.44	1.07	0.10	8.78	1
Apple M4 Max	M07	test	16,384	84.86	85.55	0.80	0.42	8.78	1
RTX 5080	M01	train	0	98.27	104.03	5.86	0.31	3.58	1
RTX 5080	M01	train	4,096	94.08	94.08	0.00	0.64	3.58	1
RTX 5080	M01	train	16,384	92.53	91.68	0.91	0.45	3.58	1
RTX 5080	M03	train	0	229.65	236.51	2.99	0.48	8.58	2
RTX 5080	M03	train	4,096	218.37	218.37	0.00	0.54	8.58	2
RTX 5080	M03	train	16,384	203.03	177.54	12.56	1.58	8.58	2
RTX 5080	M08	train	0	202.01	241.17	19.39	0.33	1.02	1
RTX 5080	M08	train	4,096	198.97	201.67	1.36	0.50	1.02	1
RTX 5080	M08	train	16,384	181.22	135.22	25.38	0.52	1.02	1
RTX 5080	M09	train	0	147.62	163.32	10.64	0.15	1.60	1

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Table 4: Protocol-complete decode measurements and B2 predictions. The retry gate uses the worst decode CV in each host–file invocation. Pre-load is the macOS one-minute POSIX load average or, on Windows, busy-logical-CPU equivalents; it is meaningful within a host but not comparable across hosts. (continued)

Host	ID	Split	Depth	Measured	B2	APE %	Row CV %	Load	Attempts
RTX 5080	M09	train	4,096	145.85	145.85	0.00	0.50	1.60	1
RTX 5080	M09	train	16,384	136.04	110.42	18.83	0.34	1.60	1
RTX 5080	M10	train	0	132.95	116.37	12.47	0.15	1.66	1
RTX 5080	M10	train	4,096	131.63	106.32	19.23	0.43	1.66	1
RTX 5080	M10	train	16,384	123.69	84.44	31.73	0.43	1.66	1
RTX 5080	M11	train	0	88.98	76.84	13.65	0.09	1.47	1
RTX 5080	M11	train	4,096	88.23	72.74	17.55	0.23	1.47	1
RTX 5080	M11	train	16,384	84.85	62.70	26.10	0.24	1.47	1
RTX 5080	M14	train	0	76.76	86.74	13.01	0.33	11.26	2
RTX 5080	M14	train	4,096	75.58	75.58	0.00	0.49	11.26	2
RTX 5080	M14	train	16,384	70.36	54.52	22.51	1.39	11.26	2
RTX 5080	M15	train	0	63.24	68.69	8.62	0.28	14.78	1
RTX 5080	M15	train	4,096	62.46	62.46	0.00	0.41	14.78	1
RTX 5080	M15	train	16,384	58.63	49.10	16.25	1.50	14.78	1
RTX 5080	M16	train	0	67.48	73.66	9.15	0.30	12.10	1
RTX 5080	M16	train	4,096	66.30	66.30	0.00	0.41	12.10	1
RTX 5080	M16	train	16,384	62.22	51.02	18.00	1.65	12.10	1
RTX 5080	M17	train	0	56.15	60.10	7.02	0.22	7.74	1
RTX 5080	M17	train	4,096	55.49	55.49	0.00	0.36	7.74	1
RTX 5080	M17	train	16,384	52.65	45.12	14.30	1.02	7.74	1
RTX 5080	M21	train	0	306.16	345.13	12.73	0.49	1.41	1
RTX 5080	M21	train	4,096	272.17	298.13	9.54	0.74	1.41	1
RTX 5080	M21	train	16,384	211.65	211.65	0.00	0.48	1.41	1
RTX 5080	M22	train	0	209.46	223.49	6.70	0.19	0.70	1
RTX 5080	M22	train	4,096	192.60	204.63	6.24	0.33	0.70	1
RTX 5080	M22	train	16,384	160.88	163.28	1.49	0.29	0.70	1
RTX 5080	M04	test	0	446.42	754.26	68.96	0.76	2.05	1
RTX 5080	M04	test	4,096	406.19	633.07	55.85	1.35	2.05	1
RTX 5080	M04	test	16,384	326.38	427.17	30.88	1.04	2.05	1
RTX 5080	M07	test	0	239.52	191.41	20.09	0.41	1.86	2
RTX 5080	M07	test	4,096	236.79	187.76	20.70	0.29	1.86	2
RTX 5080	M07	test	16,384	223.15	177.61	20.41	1.01	1.86	2

5 All scored prefill observations

These are the 99 rows exported by `results/predictions_prefill.csv`. P1 has one host coefficient; P2 has one host-by-quantization coefficient. Both are fit only at depth 0. The row CV column is reported rather than used as an inclusion filter.

Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics.

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
Apple M4 Max	M01	train	0	**primary**	589.73	547.32	7.19	589.73	0.00	0.05
Apple M4 Max	M01	train	4,096	diagnostic	491.90	547.32	11.27	589.73	19.89	1.01
Apple M4 Max	M01	train	16,384	diagnostic	382.76	547.32	42.99	589.73	54.07	3.47
Apple M4 Max	M02	train	0	**primary**	1418.29	1704.93	20.21	1610.29	13.54	0.38
Apple M4 Max	M02	train	4,096	diagnostic	1145.69	1704.93	48.81	1610.29	40.55	2.81
Apple M4 Max	M02	train	16,384	diagnostic	741.02	1704.93	130.08	1610.29	117.31	2.63
Apple M4 Max	M03	train	0	**primary**	1589.84	1556.35	2.11	1589.84	0.00	0.27
Apple M4 Max	M03	train	4,096	diagnostic	1372.89	1556.35	13.36	1589.84	15.80	2.79
Apple M4 Max	M03	train	16,384	diagnostic	848.96	1556.35	83.33	1589.84	87.27	0.72
Apple M4 Max	M08	train	0	**primary**	1463.56	1549.58	5.88	1463.56	0.00	0.07
Apple M4 Max	M08	train	4,096	diagnostic	1310.45	1549.58	18.25	1463.56	11.68	0.40
Apple M4 Max	M08	train	16,384	diagnostic	941.78	1549.58	64.54	1463.56	55.40	0.89
Apple M4 Max	M10	train	0	**primary**	819.16	727.86	11.14	687.46	16.08	0.13
Apple M4 Max	M10	train	4,096	diagnostic	727.07	727.86	0.11	687.46	5.45	0.39
Apple M4 Max	M10	train	16,384	diagnostic	564.60	727.86	28.92	687.46	21.76	1.59
Apple M4 Max	M12	train	0	**primary**	1414.02	1886.30	33.40	1781.59	25.99	0.67
Apple M4 Max	M12	train	4,096	diagnostic	1189.17	1886.30	58.62	1781.59	49.82	1.38
Apple M4 Max	M12	train	16,384	diagnostic	880.48	1886.30	114.24	1781.59	102.34	1.39
Apple M4 Max	M14	train	0	**primary**	239.52	242.31	1.17	239.52	0.00	2.73
Apple M4 Max	M14	train	4,096	diagnostic	196.04	242.31	23.60	239.52	22.18	0.31
Apple M4 Max	M14	train	16,384	diagnostic	158.60	242.31	52.78	239.52	51.02	1.33
Apple M4 Max	M15	train	0	**primary**	238.54	238.54	0.00	238.54	0.00	2.42
Apple M4 Max	M15	train	4,096	diagnostic	203.40	238.54	17.28	238.54	17.28	1.81
Apple M4 Max	M15	train	16,384	diagnostic	172.91	238.54	37.96	238.54	37.96	1.74
Apple M4 Max	M16	train	0	**primary**	235.14	238.54	1.45	235.14	0.00	1.72
Apple M4 Max	M16	train	4,096	diagnostic	207.61	238.54	14.90	235.14	13.26	2.63
Apple M4 Max	M16	train	16,384	diagnostic	180.12	238.54	32.43	235.14	30.54	1.42
Apple M4 Max	M17	train	0	**primary**	241.51	238.54	1.23	241.51	0.00	2.59
Apple M4 Max	M17	train	4,096	diagnostic	202.71	238.54	17.67	241.51	19.14	0.93
Apple M4 Max	M17	train	16,384	diagnostic	164.64	238.54	44.89	241.51	46.69	2.76
Apple M4 Max	M18	train	0	**primary**	224.91	238.54	6.06	224.91	0.00	3.91
Apple M4 Max	M18	train	4,096	diagnostic	191.29	238.54	24.70	224.91	17.58	0.80

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Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics. (continued)

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
Apple M4 Max	M18	train	16,384	diagnostic	162.37	238.54	46.91	224.91	38.52	2.61
Apple M4 Max	M19	train	0	**primary**	225.20	238.54	5.92	225.20	0.00	2.87
Apple M4 Max	M19	train	4,096	diagnostic	196.00	238.54	21.70	225.20	14.90	2.15
Apple M4 Max	M19	train	16,384	diagnostic	174.79	238.54	36.47	225.20	28.84	1.41
Apple M4 Max	M20	train	0	**primary**	239.03	238.54	0.20	239.03	0.00	2.54
Apple M4 Max	M20	train	4,096	diagnostic	208.07	238.54	14.64	239.03	14.88	2.22
Apple M4 Max	M20	train	16,384	diagnostic	181.38	238.54	31.52	239.03	31.78	1.35
Apple M4 Max	M21	train	0	**primary**	2135.94	2119.33	0.78	2001.68	6.29	0.04
Apple M4 Max	M21	train	4,096	diagnostic	1653.76	2119.33	28.15	2001.68	21.04	2.82
Apple M4 Max	M21	train	16,384	diagnostic	965.07	2119.33	119.60	2001.68	107.41	1.34
Apple M4 Max	M22	train	0	**primary**	2239.31	2119.33	5.36	2239.31	0.00	0.12
Apple M4 Max	M22	train	4,096	diagnostic	1735.34	2119.33	22.13	2239.31	29.04	0.12
Apple M4 Max	M22	train	16,384	diagnostic	892.27	2119.33	137.52	2239.31	150.97	1.34
Apple M4 Max	M04	test	0	**primary**	4208.20	4547.98	8.07	4295.53	2.08	0.42
Apple M4 Max	M04	test	4,096	diagnostic	3300.39	4547.98	37.80	4295.53	30.15	0.62
Apple M4 Max	M04	test	16,384	diagnostic	1848.16	4547.98	146.08	4295.53	132.42	1.94
Apple M4 Max	M05	test	0	**primary**	236.65	233.97	1.13	220.60	6.78	3.52
Apple M4 Max	M05	test	4,096	diagnostic	187.15	233.97	25.01	220.60	17.87	3.95
Apple M4 Max	M05	test	16,384	diagnostic	176.87	233.97	32.28	220.60	24.72	1.62
Apple M4 Max	M06	test	0	**primary**	1203.59	1820.39	51.25	1719.34	42.85	0.45
Apple M4 Max	M06	test	4,096	diagnostic	1106.68	1820.39	64.49	1719.34	55.36	3.64
Apple M4 Max	M06	test	16,384	diagnostic	895.52	1820.39	103.28	1719.34	91.99	1.19
Apple M4 Max	M07	test	0	**primary**	1259.18	1640.13	30.25	1549.08	23.02	4.84
Apple M4 Max	M07	test	4,096	diagnostic	1200.41	1640.13	36.63	1549.08	29.05	3.30
Apple M4 Max	M07	test	16,384	diagnostic	995.75	1640.13	64.71	1549.08	55.57	1.37
RTX 5080	M01	train	0	**primary**	5624.55	4619.99	17.86	5624.55	0.00	5.17
RTX 5080	M01	train	4,096	diagnostic	5030.69	4619.99	8.16	5624.55	11.80	3.92
RTX 5080	M01	train	16,384	diagnostic	4336.80	4619.99	6.53	5624.55	29.69	2.60
RTX 5080	M03	train	0	**primary**	8633.84	13137.34	52.16	8633.84	0.00	3.75
RTX 5080	M03	train	4,096	diagnostic	8160.96	13137.34	60.98	8633.84	5.79	4.00
RTX 5080	M03	train	16,384	diagnostic	7086.04	13137.34	85.40	8633.84	21.84	2.75
RTX 5080	M08	train	0	**primary**	9911.27	13080.14	31.97	12128.70	22.37	5.67
RTX 5080	M08	train	4,096	diagnostic	9373.61	13080.14	39.54	12128.70	29.39	4.03
RTX 5080	M08	train	16,384	diagnostic	8143.16	13080.14	60.63	12128.70	48.94	3.62
RTX 5080	M09	train	0	**primary**	10379.35	13080.14	26.02	13011.54	25.36	5.85
RTX 5080	M09	train	4,096	diagnostic	9802.11	13080.14	33.44	13011.54	32.74	4.92
RTX 5080	M09	train	16,384	diagnostic	8448.91	13080.14	54.81	13011.54	54.00	3.69

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Table 5: Protocol-complete prefill measurements and predictions. The retry decision was decode-gated, not prefill-gated; depth-0 rows are the primary evaluation and larger depths are scope diagnostics. (continued)

Host	ID	Split	Depth	Scope	Measured	P1	P1 APE	P2	P2 APE	Row CV %
RTX 5080	M10	train	0	**primary**	6452.64	6143.96	4.78	5697.05	11.71	3.37
RTX 5080	M10	train	4,096	diagnostic	6219.18	6143.96	1.21	5697.05	8.40	2.66
RTX 5080	M10	train	16,384	diagnostic	5738.49	6143.96	7.07	5697.05	0.72	2.17
RTX 5080	M11	train	0	**primary**	6861.31	6143.96	10.45	6111.74	10.92	4.42
RTX 5080	M11	train	4,096	diagnostic	6647.95	6143.96	7.58	6111.74	8.07	2.90
RTX 5080	M11	train	16,384	diagnostic	6079.76	6143.96	1.06	6111.74	0.53	2.86
RTX 5080	M14	train	0	**primary**	1725.38	2045.35	18.54	1725.38	0.00	3.23
RTX 5080	M14	train	4,096	diagnostic	1682.57	2045.35	21.56	1725.38	2.54	1.97
RTX 5080	M14	train	16,384	diagnostic	1538.25	2045.35	32.97	1725.38	12.17	2.17
RTX 5080	M15	train	0	**primary**	2024.12	2013.56	0.52	2024.12	0.00	3.50
RTX 5080	M15	train	4,096	diagnostic	1954.57	2013.56	3.02	2024.12	3.56	2.46
RTX 5080	M15	train	16,384	diagnostic	1771.53	2013.56	13.66	2024.12	14.26	2.22
RTX 5080	M16	train	0	**primary**	2043.07	2013.56	1.44	2043.07	0.00	3.11
RTX 5080	M16	train	4,096	diagnostic	1962.09	2013.56	2.62	2043.07	4.13	3.14
RTX 5080	M16	train	16,384	diagnostic	1783.83	2013.56	12.88	2043.07	14.53	2.23
RTX 5080	M17	train	0	**primary**	2042.02	2013.56	1.39	2042.02	0.00	17.04
RTX 5080	M17	train	4,096	diagnostic	2114.20	2013.56	4.76	2042.02	3.41	2.90
RTX 5080	M17	train	16,384	diagnostic	1907.04	2013.56	5.59	2042.02	7.08	2.39
RTX 5080	M21	train	0	**primary**	16588.19	17889.45	7.84	16588.19	0.00	7.30
RTX 5080	M21	train	4,096	diagnostic	14035.04	17889.45	27.46	16588.19	18.19	5.06
RTX 5080	M21	train	16,384	diagnostic	9948.68	17889.45	79.82	16588.19	66.74	3.46
RTX 5080	M22	train	0	**primary**	17795.63	17889.45	0.53	17795.63	0.00	8.37
RTX 5080	M22	train	4,096	diagnostic	14813.44	17889.45	20.76	17795.63	20.13	5.96
RTX 5080	M22	train	16,384	diagnostic	10330.65	17889.45	73.17	17795.63	72.26	3.83
RTX 5080	M04	test	0	**primary**	11706.91	38390.00	227.93	35597.53	204.07	39.19
RTX 5080	M04	test	4,096	diagnostic	13303.93	38390.00	188.56	35597.53	167.57	5.45
RTX 5080	M04	test	16,384	diagnostic	10873.99	38390.00	253.04	35597.53	227.36	4.90
RTX 5080	M07	test	0	**primary**	11433.58	13844.48	21.09	12837.44	12.28	4.86
RTX 5080	M07	test	4,096	diagnostic	10797.04	13844.48	28.22	12837.44	18.90	3.44
RTX 5080	M07	test	16,384	diagnostic	9215.27	13844.48	50.23	12837.44	39.31	3.36

6 Repeatability and calibration

A separate repeatability series exists only for Apple M4 Max; RTX 5080 variability is represented by the five within-invocation repetitions in the row tables, not by an equivalent across-run series. The six Mac runs give mean decode throughput 166.457 tokens/s and sample between-run CV 1.613%; prefill is 2133.091 tokens/s with CV 0.373%. The max-minus-min spans are 4.34% and 0.97% of their respective means. Decode has a descriptive linear trend of 0.74% per run. These are single-cell diagnostics, not general detection thresholds.

Run	Timestamp	Settle (s)	Prefill	Decode	Decode within-run SD
1	2026-08-30T01:06:08.096930-07:00	45	2137.572	164.323	6.520
2	2026-08-30T01:06:59.837356-07:00	45	2136.650	165.447	1.473
3	2026-08-30T01:07:51.671530-07:00	45	2136.742	163.200	10.629
4	2026-08-30T01:08:43.573411-07:00	45	2116.939	166.925	2.338
5	2026-08-30T01:09:35.318847-07:00	45	2135.238	168.415	0.805
6	2026-08-30T01:10:26.991582-07:00	45	2135.403	170.432	0.464

The append-only Mac measurement file also preserves a complete pre-protocol decode grid. It is excluded from every fit and score above. The comparison is Mac-only and is not pooled with RTX data.

Pass	All *n*	Test *n*	Test MAPE	Median	P90	Max	All-row CV med.	All-row CV max
Archived pre-protocol	57	12	19.30	15.77	25.94	74.59	1.74	14.91
Final Mac protocol-complete	57	12	13.11	8.98	36.65	49.92	1.00	3.03

For comparison, the retained historical uncontrolled file contains two depth-0 observations of the same Qwen3.5-4B Q4_K_M cell:

Phase	Higher throughput	Lower throughput	Decrease (%)
prefill	1467.424	1090.676	25.67
decode	103.616	87.594	15.46

6.1 Host settings and calibration

Host	Platform	Accelerator	Threads	Reps	Settle (s)	Requested ngl
Apple M4 Max	Darwin-arm64	arm (Metal)	12	5	45	99
RTX 5080	Windows-AMD64	NVIDIA GeForce RTX 5080	16	5	45	99

The requested `n_gpu_layers=99` setting asks the runtime to offload as many layers as it can; it is not direct telemetry proving that every tensor and KV allocation remained resident on the accelerator. No device-memory trace was recorded, so the appendix does not label these observations as confirmed fully resident.

Host	Calibration	Value	Detail
Apple M4 Max	CPU STREAM-style triad	48.558 GB/s	7 repetitions
Apple M4 Max	CPU FP32 matrix multiply	2.458 TFLOP/s	5 repetitions
Apple M4 Max	MPS device copy	380.052 GB/s	arm (Metal)
Apple M4 Max	MPS FP16 matrix multiply	13.950 TFLOP/s	arm (Metal)
Apple M4 Max	LLM reference ceiling	377.244 GB/s	`Qwen3.5-9B-Q8_0.gguf`
RTX 5080	CPU STREAM-style triad	13.458 GB/s	7 repetitions
RTX 5080	CPU FP32 matrix multiply	2.314 TFLOP/s	5 repetitions
RTX 5080	CUDA device copy	801.061 GB/s	NVIDIA GeForce RTX 5080
RTX 5080	CUDA FP16 matrix multiply	118.829 TFLOP/s	NVIDIA GeForce RTX 5080

For the fitted B1/B2 and P1/P2 models, the measured bandwidth or FLOP/s scale cancels algebraically after fitting on this same host. It sets the scale of B0 and supports roofline interpretation; it does not by itself produce the reported fitted-model accuracy. The exported row-level bandwidth field uses the device-copy result on both hosts. The Mac LLM reference is retained as a diagnostic and is not the B0 bandwidth value used in the current exports.

Host	Probe	File GB	Decode	SD	CV (%)	Effective GB/s
Apple M4 Max	`Qwen3.8-27B-Q8_0.gguf`	29.05	7.100	4.199	59.14	196.653
Apple M4 Max	`Qwen3.5-9B-Q8_0.gguf`	9.53	44.661	0.586	1.31	377.244
Apple M4 Max	`Qwen3.5-4B-Q8_0.gguf`	4.48	70.817	5.524	7.80	317.432

7 Output-projection sensitivity

The additive output-projection term is reported as a sensitivity analysis. It is not selected: held-out improvement is not consistent across loss functions and hosts, and the term partly recharges weights already included in the streamed-byte term. Host rows and the unequally weighted all-host aggregate are shown separately. The one-term rows in this section are refit under the listed time-domain losses; they are not the median-ratio B2 headline and need not reproduce its MAPE.

Host	Loss	Model	Split	*n*	MAPE	Median	P90	Max
Apple M4 Max	absolute time	one term	train	45	12.38	4.97	34.07	91.71
Apple M4 Max	absolute time	one term	test	12	19.26	14.43	46.77	61.03
Apple M4 Max	absolute time	two terms	train	45	9.88	4.47	29.64	62.39
Apple M4 Max	absolute time	two terms	test	12	34.03	30.21	62.96	72.96
RTX 5080	absolute time	one term	train	36	14.18	11.77	28.14	45.36
RTX 5080	absolute time	one term	test	6	44.01	31.40	97.73	105.71
RTX 5080	absolute time	two terms	train	36	12.84	9.76	30.05	39.86
RTX 5080	absolute time	two terms	test	6	27.30	24.89	55.35	56.39
all hosts	absolute time	one term	train	81	13.18	8.06	33.36	91.71
all hosts	absolute time	one term	test	18	27.51	14.43	69.65	105.71
all hosts	absolute time	two terms	train	81	11.20	5.90	30.18	62.39
all hosts	absolute time	two terms	test	18	31.79	30.21	61.58	72.96
Apple M4 Max	relative time	one term	train	45	11.40	4.35	28.13	83.22
Apple M4 Max	relative time	one term	test	12	15.17	11.87	40.27	53.89
Apple M4 Max	relative time	two terms	train	45	8.34	4.68	18.01	46.86
Apple M4 Max	relative time	two terms	test	12	20.58	19.27	38.22	45.61
RTX 5080	relative time	one term	train	36	12.55	11.72	22.83	34.79
RTX 5080	relative time	one term	test	6	40.81	29.12	83.35	90.75
RTX 5080	relative time	two terms	train	36	11.80	10.35	20.95	28.62
RTX 5080	relative time	two terms	test	6	28.48	23.66	54.83	57.34
all hosts	relative time	one term	train	81	11.91	8.81	26.26	83.22
all hosts	relative time	one term	test	18	23.72	11.87	60.51	90.75
all hosts	relative time	two terms	train	81	9.87	6.65	20.34	46.86
all hosts	relative time	two terms	test	18	23.22	19.27	47.62	57.34

8 Provenance and explicit limits

The on-disk inputs used for this appendix have the following SHA-256 digests. The repository HEAD at generation was `1bdce5e0d917dce152cba4809d05fc1922a98797`; the generator does not assert a clean working tree, so the file digests, rather than this commit alone, identify the inputs.

Table 6: Exact inputs used to generate this appendix.

Role	Path	SHA-256
measurements (lun-mac)	`results/measurements_lun-mac.csv`	`99dcbf9abb9454877713198909703c1f77e00555f0258f7bf9d83e4395be99b4`
measurements (rtx5080)	`results/measurements_rtx5080.csv`	`5f4d46f4952b2dadc923628c5256edb8adff77301aac835d252dd07b17393e91`
decode predictions	`results/predictions.csv`	`5c0020859871a943a960f1bcf7c96e40bc7abd2005dd817c2fcb9962af6f4066`
prefill predictions	`results/predictions_prefill.csv`	`430ca9a302b263c8bec7aae51bfb9d2bbf79182c1c4d9c9c1efbd6712ee9e436`
decode aggregate error table	`results/error_table.csv`	`37f654f71d160a3c4676fe685f1a36335998fdd89d317f29e314fcc0975fa4af`
decode error table by host	`results/error_table_by_host.csv`	`4dfb6874a0f3fd33bb4d2c747ed4da8d8ff916314003658885304299109f158a`
prefill aggregate error table	`results/error_table_prefill.csv`	`067f16c7aa5677a234cf59dfe6fccfc967d62e8348b63014adcfcf952485c2b5`
prefill error table by host	`results/error_table_prefill_by_host.csv`	`a5a1b9ba99396f7571b01617b381fdbe1fb8d063c07a42913d633230d5e56692`
calibration (lun-mac)	`results/calibration_lun-mac.json`	`1848525fd55119b745d16c51a53f0fa7793c81c6d5c4738af38bb838795ee5a8`
calibration (rtx5080)	`results/calibration_rtx5080.json`	`4a23505686da0f75a594fef75ac8047c5f06ff63b2bbd43856c0e18cde631591`
environment (lun-mac)	`results/env_lun-mac.json`	`1cda40d02b90d0ca21ab7c9cfc02cbf6c36bf920a74168c88a78a1456ab7ef59`
environment (rtx5080)	`results/env_rtx5080.json`	`0f695cc20ca99843c58f656e05303377f11d4083f357beadb7da8d72a02c7f9f`
manifest	`results/model_manifest.json`	`c24b4d8bfb4c1b27a4204775269e8e02f46936914e4f72117f1c955f600ff5c7`
metadata	`results/model_metadata.json`	`cb1afadb7fad3b6d9d2df88b7176d72f4c25068bbb865e8c9a33cc79a3e4a2e5`
repeatability (lun-mac)	`results/repeatability_lun-mac.csv`	`f79f9565aebc662bde98d1fefa44b29d73e83188c60b6bf60a1061cc2de9f622`
archived uncontrolled comparison	`results/contaminated/measurements_UNCONTROLLED_lun-mac.csv.bak`	`b344118d3902a9aa6530a6113a3c5d0c0310ae2a703ad4986c8146d70821eb50`
analysis implementation	`llmperf/analyze.py`	`f94a0b83f4d8cd5a3c7e1bf8da3a8b4121ecd3578191484b8b7e7c39bb6a5374`
refinement implementation	`llmperf/refine.py`	`523dd6972e20543c314a4f40761ce5fd75da5d2f6b86532c739d022fed535786`
metadata implementation	`llmperf/common.py`	`f1f4a7cd9a927b05344cec77ebad6492b8f8195509f897df5443ca50604df592`
appendix generator	`paper/icassp2027/generate_supplement.py`	`be7452a1e12d4e9ec1b3f980f200c1c4e8339ce1d456f3adce3f3cb3cd59a208`

- Apple M4 Max and RTX 5080 contributed unequal, overlapping cohorts. Headline coefficients are fit separately by host. The leave-one-host-out audit uses measured target bandwidth and only two systems, so it does not establish universal transfer to new runtime stacks or formats.
- Every scored row records a requested `n_gpu_layers=99`, but no device-memory trace or runtime-reported resident-layer count was retained. This is a maximal-offload request, not proof of full accelerator residency; no partial-offload sweep or offload-cliff result exists.
- The largest RTX modeled file-plus-KV footprint is 17.508 GB for `Qwen3.8-27B-UD-Q3_K_XL.gguf` at depth 16,384, while calibration reports 17.094 GB of device memory. The differing accounting conventions further preclude a residency claim.
- The Mac rows use the placeholder commit value `nogit`; both runtime version queries returned usage lines rather than immutable `llama.cpp` revisions. The RTX project commit field is not itself a runtime binary revision.
- Manifest entries identify repository, filename, and byte size, but not immutable repository revisions or full-file hashes. Exact experimental replay is therefore not guaranteed.
- Derived parameter counts and per-depth KV bytes are consumed from the frozen metadata snapshot during appendix generation. This reproduces the analysis but is not a fresh, independent metadata extraction.
- Held-out coverage is small: 4 Mac and 2 RTX configurations, all Q4-family. Other quantization formats occur only in training data. No learning curve establishes how many reference configurations suffice.
- Decode MAPE treats three depths from each host–file configuration as observations; those rows are correlated. No narrow confidence claim is warranted.
- Prefill's primary result is restricted to depth 0. Larger-depth predictions are scope diagnostics because the equation has no existing-prefix term. The retry rule is decode-only, and 29 of 42 scored RTX prefill rows exceed 3% within-run CV; the RTX prefill result should therefore be treated as both inaccurate and noisy.
- The three Q8 calibration probes are excluded only from the Mac fit and score; two corresponding files are scored on RTX. The 63.39-GB gpt-oss-120B Mac attempt is recorded as a prompt-batch decode failure and contributes no throughput row; it was not attempted on RTX.
- Qwen3.8-27B IQ2 has 64 layers and 26.90B parameters, whereas the other shown Qwen3.8-27B files have 65 layers and 27.32B parameters. The quantization plot is therefore a near-family comparison, not a controlled bit-format substitution for one identical network.

The metadata snapshot additionally records these parser/source Git object IDs:

Object	Identifier
Code commit	`1bdce5e0d917dce152cba4809d05fc1922a98797`
Measurement blob	`0ec72c61f2eaa371ba4f591a28964ec602d0686b`
Predictions blob	`de86a4286026e3b895b3c3809f9b4b0eb7dbda47`
Shape-log blob	`fa3d7b9cbf4a731219c7d52829fde031c67e3a2d`